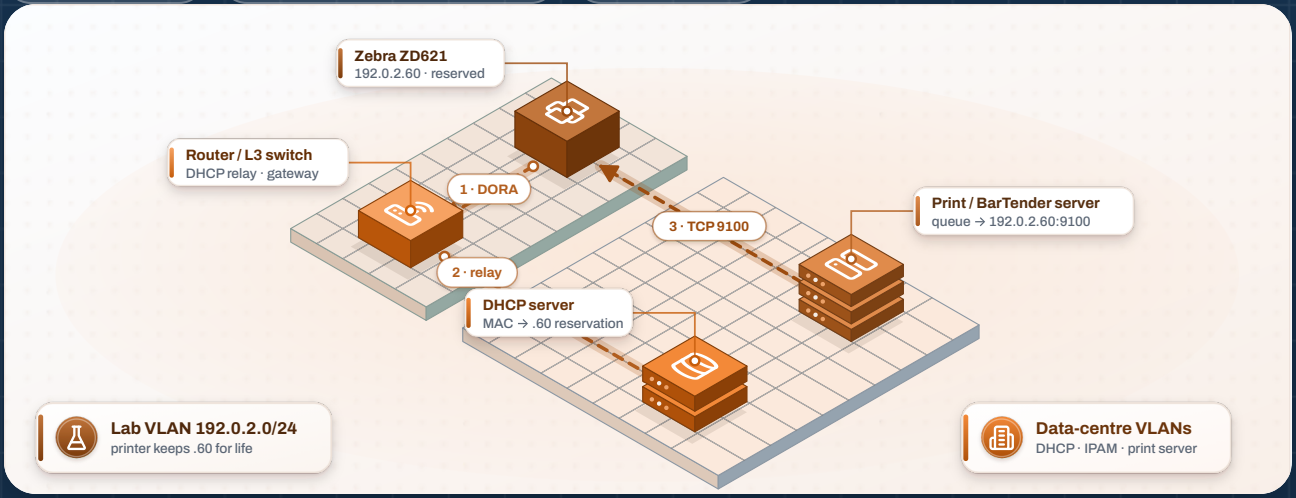
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-LBL-002

Static IP Reservation and Network Label Printer Integration

Give a network label printer a permanent address and connect it to Windows, BarTender and the LIMS, four ways: DHCP console reservation, a DhcpServer PowerShell template, static configuration on the printer, and scripted queue integration with Add-PrinterPort / Add-Printer.

Label Printing | Networking · DHCP / IPAM | Lab Systems



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Sanitized reference edition — client identifiers removed

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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain the DHCP DORA exchange, relays, leases and how a reservation ties an address to a MAC address.

- Choose between a DHCP reservation, a static address configured on the printer and an IPAM-managed request, and know the risks of each.
- Create, correct and clean up reservations and stale leases in the DHCP console and with the DhcpServer PowerShell module.
- Diagnose an unreachable printer layer by layer with the configuration label, ping, ARP and Test-NetConnection, and recognise MAC-binding conflicts.
- Integrate the printer into a Windows print queue, BarTender and the LIMS, and verify end to end.
- Escalate to Networking or IPAM with evidence that gets the problem solved on the first pass.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
≈ 30–60 min (plus approval lead time)	Intermediate	DHCP Administrators (or IPAM request) + print-server admin	Low · one device; address conflicts possible

CHAPTER 1

Concepts You Need First

Network label printers sit at the junction of three teams: the lab that uses them, the application team that owns BarTender and the LIMS, and the Networking or IP Address Management (IPAM) team that owns addresses. In the reference engagement a new Zebra ZD621 was given a "static IP", yet it stayed unreachable for days. The address had previously been bound to the MAC address of a retired printer, the DHCP server kept that stale binding, and every team's individual check came back "looks fine". This SOP teaches the networking ideas behind that failure, gives four ways to do the job correctly, and shows how to escalate with evidence that ends the back-and-forth.

1.1 Why a printer needs a permanent address

Workstations can change IP address every day without anyone noticing, because people reach them by name. Network printers are different: the Windows print queue on the print server points at a **TCP/IP port** holding an IP address, and BarTender and the LIMS print through that queue. If the printer's address changes, the queue sends jobs into the void. A label printer therefore needs an address that never changes, and that the network team knows about so it is never handed to anything else.

KEY CONCEPT · RESERVATION VERSUS STATIC

There are two ways to make an address permanent. A **DHCP reservation** keeps the printer on DHCP but tells the server "always give this MAC address this IP". A **static (on-device) address** is typed into the printer itself and DHCP is turned off. A reservation is like a named seat at a restaurant: the host still seats you, but always at your table. A static address is like bringing your own chair: it works until the restaurant rearranges the room.

Approach	Strengths	Risks
DHCP reservation	Central record; gateway, mask and DNS come from the scope; survives moves within the scope; easy to audit with PowerShell.	Bound to the MAC address: replacing the printer or its network card silently breaks it until the reservation is updated.
Static on device	Works with no DHCP at all (isolated instrument networks); no dependency on a server.	Invisible to DHCP unless excluded from the pool; wrong after a move to another subnet; easy to duplicate.

IPAM-managed request	One source of truth (Windows IPAM, Infoblox and similar); conflict checks built in.	Lead time and a hand-off to another team; still realised as a reservation or static address.

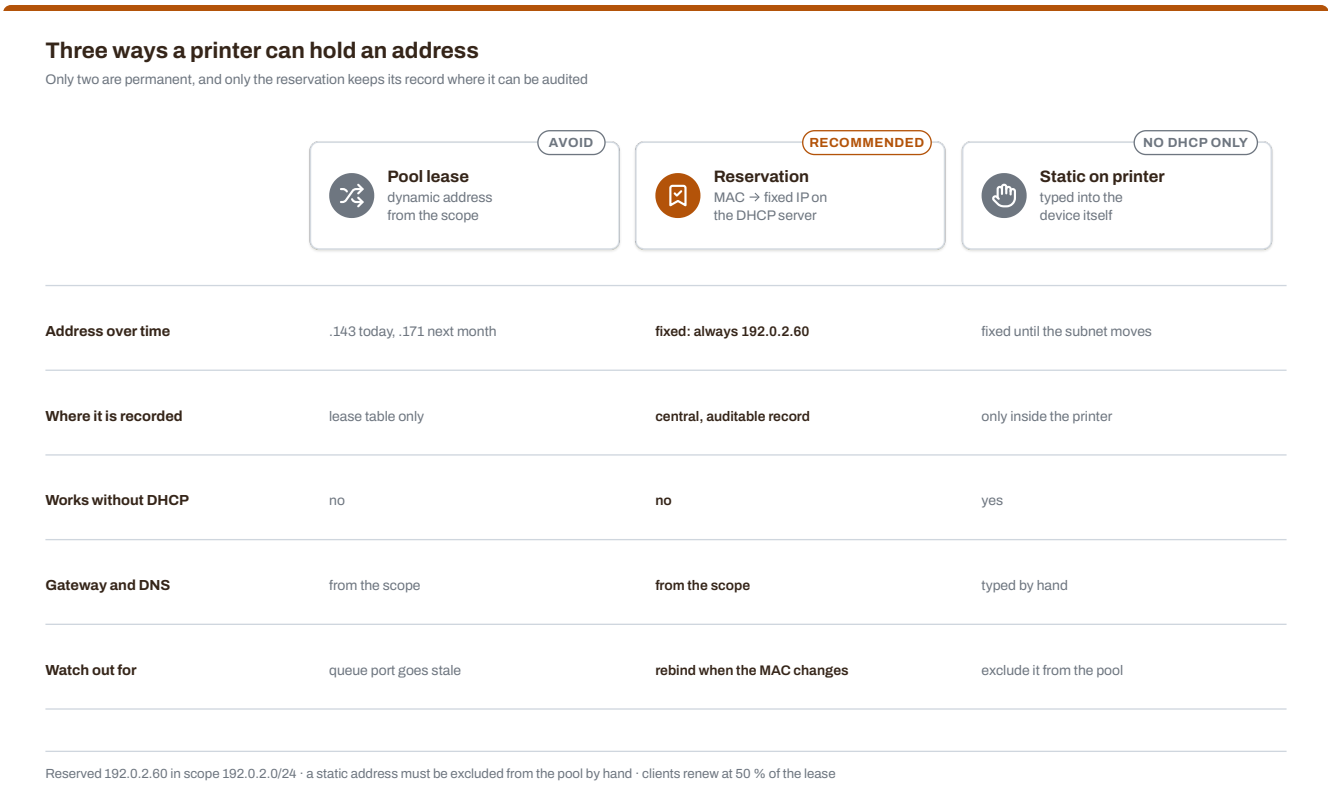


Figure 1. Use a DHCP reservation for every print queue target; a static address is the fallback only where no DHCP exists, and a pool lease is never acceptable.

1.2 How DHCP hands out an address: DORA

The Dynamic Host Configuration Protocol (DHCP) uses a four-message exchange remembered as **DORA**: Discover, Offer, Request, Acknowledge. A client with no address broadcasts a *Discover*. Broadcasts do not cross routers, so on enterprise networks the router or layer-3 switch acts as a **DHCP relay** (Cisco calls it `ip helper-address`): it forwards the request to the central DHCP server and stamps it with its own interface address (*giaddr*), which tells the server which scope to use.

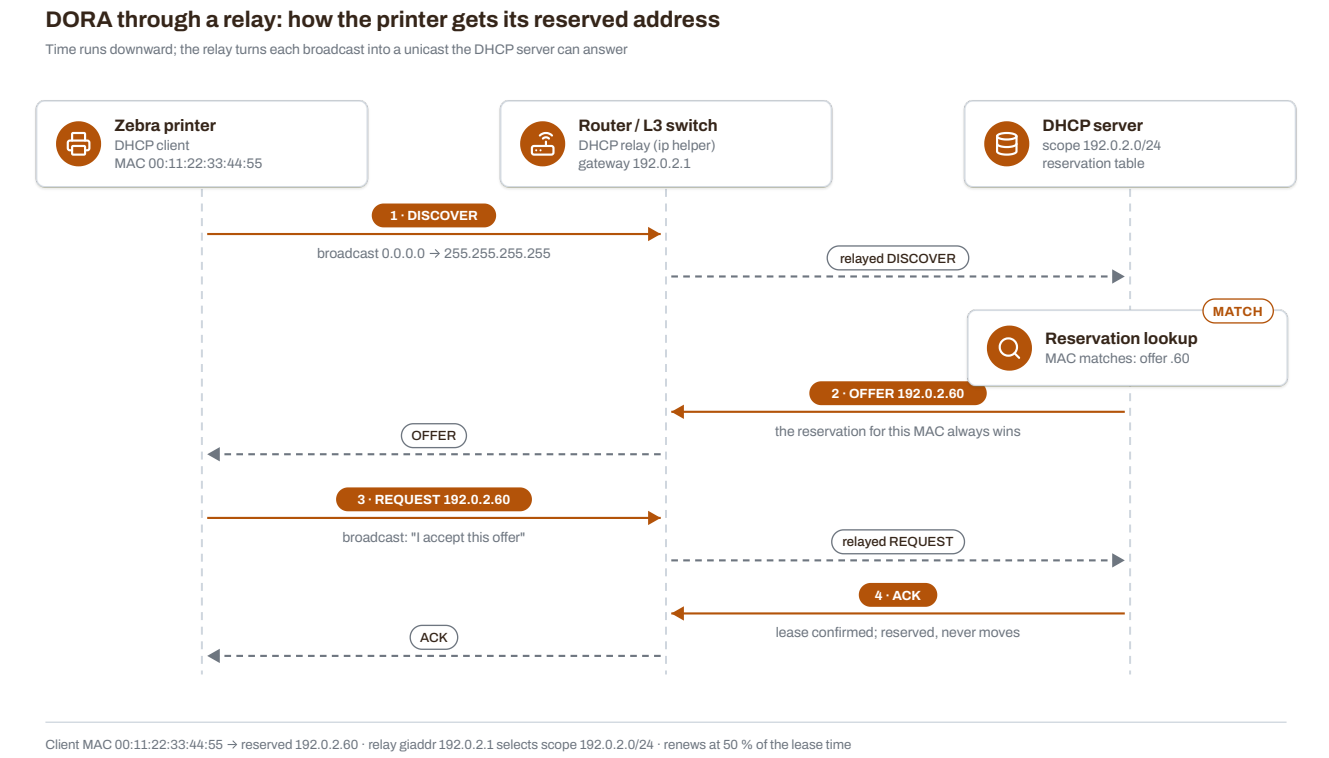


Figure 2. Because the server matches the client MAC before it looks at the pool, a correct reservation is always offered in preference to any free address.

- **Scope**: the range of addresses for one subnet (e.g. 192.0.2.0/24), plus options such as router (gateway), DNS servers and lease time.
- **Pool and exclusions**: the part of the scope handed out dynamically; exclusions carve out addresses reserved for static devices.
- **Lease**: the server's record that an address is in use by a client, with an expiry time. Clients renew at half the lease time.
- **Reservation**: a permanent mapping of client identifier (the MAC address for printers) to one IP address.

WHY IT MATTERS · LEASES OUTLIVE HARDWARE

When a printer is replaced, its old lease and reservation do not vanish. They stay until someone deletes them or the lease expires. The new printer arrives with a new MAC address, finds no matching reservation, and quietly receives a random pool address, or none at all if the pool is full. Meanwhile the print queue still points at the reserved address, where nothing answers.

1.3 MAC binding conflicts and stale leases

This is exactly what happened in the reference case. The requested address had been reserved years earlier for another printer's MAC. The new printer was correctly set to DHCP, but the server would not give it the reserved address because the reservation named a different MAC; and even after the reservation was rebound, the existing lease and the printer's own cached state delayed the change until the stale lease was removed and the printer renewed.

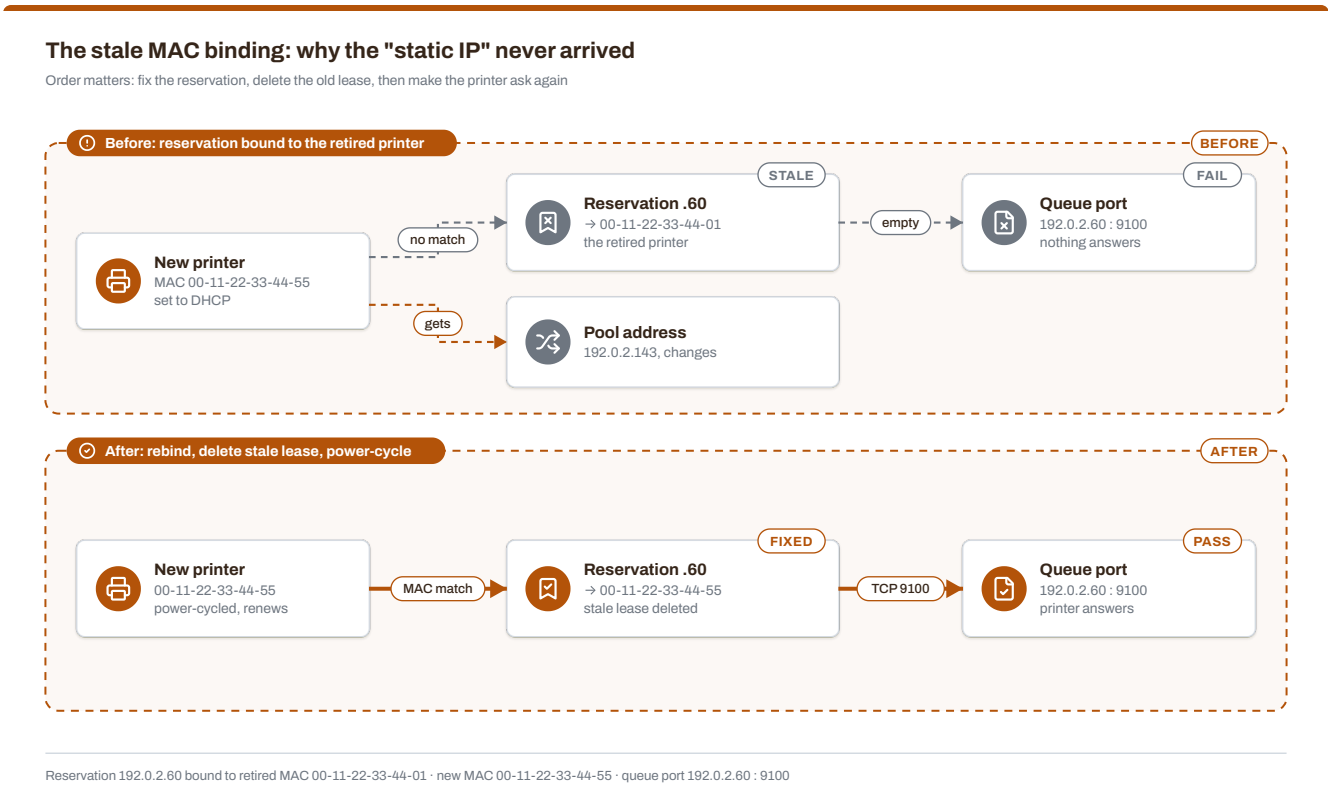


Figure 3. Replacing a printer changes its MAC, so the reservation must be rebound and the old lease removed before the new device will receive the address.

1.4 ARP: how a device is found on the local segment

An IP address is only a logical label. On the local network, frames are delivered by MAC address, and the **Address Resolution Protocol (ARP)** translates one into the other: "who has 192.0.2.60? Tell 192.0.2.10". The device that owns the address replies with its MAC. Your workstation and the router keep an **ARP cache** of recent answers, and those caches are powerful evidence:

- *No ARP entry* (or *Incomplete*): nothing on the segment is answering for that IP. The printer does not hold the address.
- *ARP entry with the expected MAC*: the printer holds the address; if ping still fails, look at firewalls or the printer's own settings.

- *ARP entry with a different MAC*: another device holds the address, an **IP conflict**. Find it before anything else.

> **NOTE · ARP ONLY WORKS WITHIN A SUBNET**
>
> From the print server in another VLAN you will only ever see the router's MAC. Ask Networking to check the ARP table on the printer's gateway, or test from a machine in the same lab VLAN.

1.5 VLANs, routing and firewalls

Lab printers commonly live in a dedicated lab or device VLAN, while the print and BarTender servers sit in a data-centre VLAN. Traffic crosses at least one router and often a firewall. For a print job to arrive, three things must be true: the printer's switch port is in the right VLAN, the router has an interface (gateway) in that subnet, and the firewall allows the print server to reach the printer on **TCP 9100** (raw printing). Management protocols such as HTTP/HTTPS for the web page and SNMP (UDP 161) for status may need separate rules.

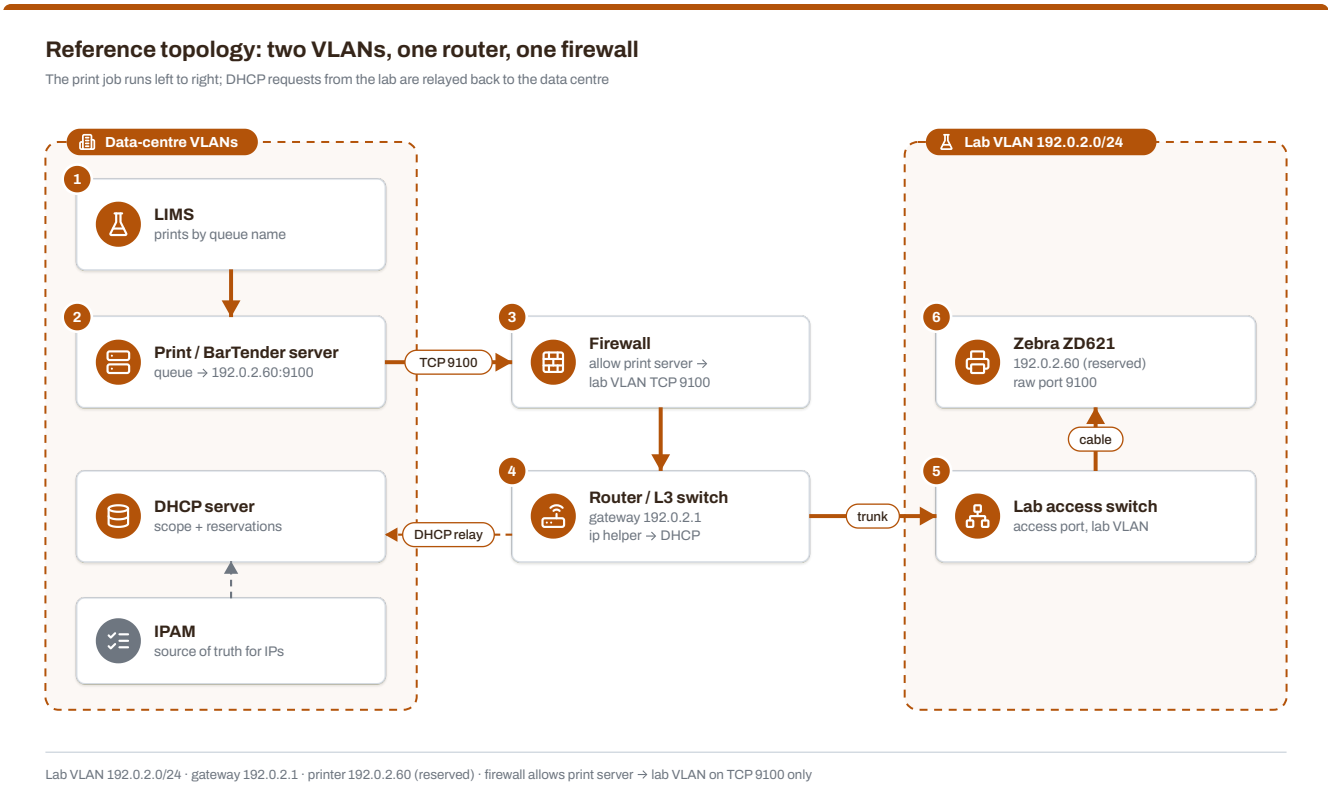


Figure 4. Print jobs and DHCP requests cross the same router in opposite directions, so a fault in either path can look like an unreachable printer.

1.6 Choosing a method

A · DHCP console

Create or correct the reservation in the DHCP MMC and clean stale leases.

BEST FOR: ONE PRINTER, DHCP OWNERS

B · DhcpServer PowerShell

Template that inspects, backs up, rebinds, cleans and replicates, with evidence.

BEST FOR: REPEATABLE, AUDITED CHANGES

C · Static on the printer

Front panel, web page or SGD commands; DHCP turned off.

BEST FOR: NO DHCP, ISOLATED NETWORKS

D · Queue + BarTender

Add-PrinterPort / Add-Printer, then BarTender and LIMS registration.

BEST FOR: EVERY METHOD FINISHES HERE

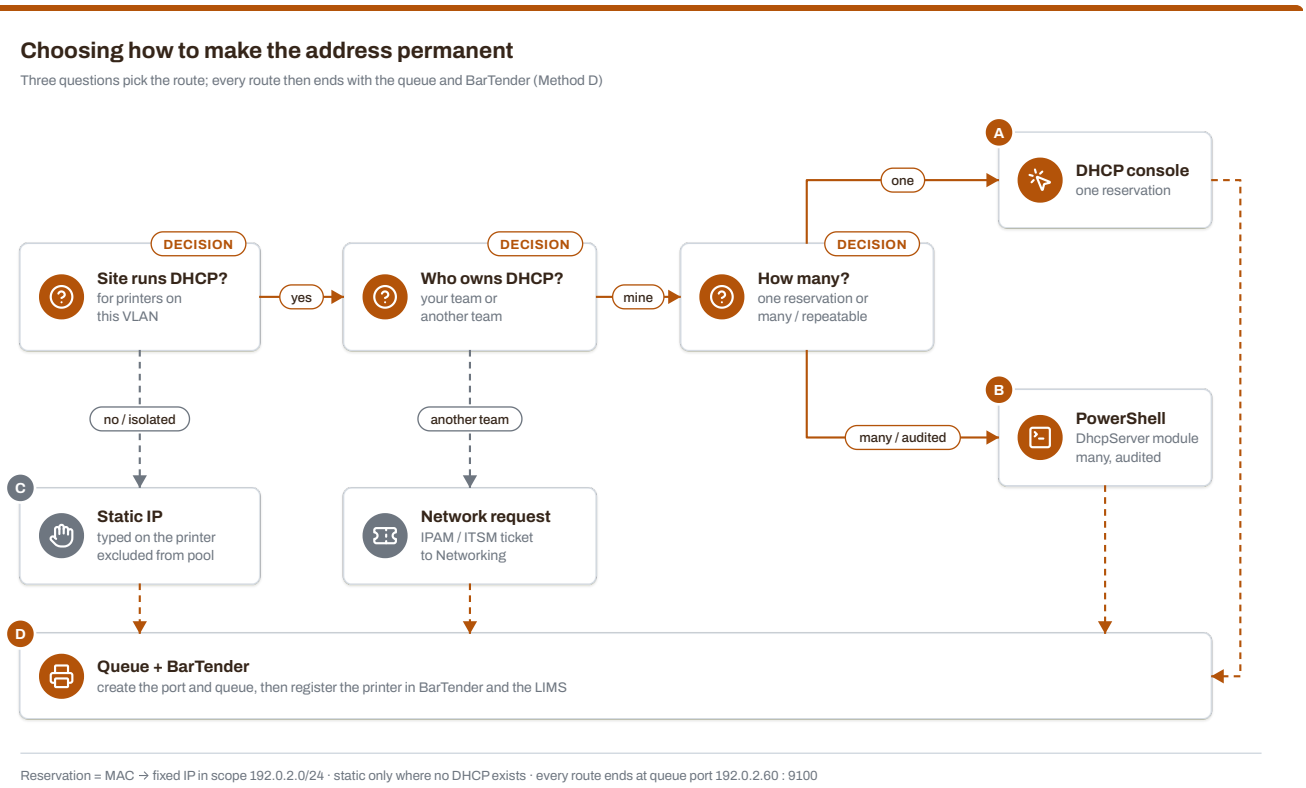


Figure 5. Who owns DHCP and how many printers are involved decide the route, but every route finishes by pointing the queue and BarTender at the address.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Task	Least-privilege access	Notes
Read scopes, leases, reservations	**DHCP Users** (local group on the DHCP server)	Enough for diagnosis with `Get-DhcpServerv4*` cmdlets.
Create / delete reservations and leases	**DHCP Administrators** on the DHCP server (or delegated domain group)	Not Domain Admins. Install RSAT DHCP tools on your admin workstation.
Same task through IPAM	Windows IPAM: *IPAM DHCP Reservations Administrator*; Infoblox and similar: permission on the network's fixed addresses	Where IPAM is the source of truth, change it there, not directly on the DHCP server.
Firewall and VLAN changes	Networking / security team via change request	You supply source, destination, port and justification.
Static IP on the printer	Physical access, or the printer's web admin password	Change default printer passwords before production.
Create port and queue	Local **Administrators** on the print / BarTender server	Driver installation is admin-only since the 2021 Point and Print hardening.
BarTender and LIMS registration	BarTender administrator; LIMS application administrator	Usually the label application team.

> **GOOD PRACTICE · TIME-BOUND PRIVILEGE**
>
> Hold DHCP Administrators and print-server admin rights only for the change window: request them through your privileged access management tool, or use a dedicated admin account whose group membership is time-limited. The approval record doubles as audit evidence.

2.2 Prerequisites checklist

- ☐ Approved request in the ITSM / ServiceNow request portal for a permanent address (or IPAM ticket), referencing the change number (e.g. `CHG0012345`).
- ☐ The printer's **MAC address**, read from its network configuration label, not from the box or a spreadsheet.

- [ ] Subnet, gateway and VLAN for the printer's switch port; confirmation that the port is patched and enabled.
- [ ] A free address *outside* the dynamic pool or an IPAM-allocated one; confirmation that no existing reservation or lease holds it.
- [ ] Firewall rule (print server → printer, TCP 9100) approved or confirmed existing.
- [ ] Correct ZPL driver for the model and resolution available on the print server (see SOP-LBL-001).
- [ ] A person at the printer for power-cycles and test labels.

CHAPTER 3

Method A · DHCP Console Reservation

The DHCP Microsoft Management Console (`dhcpmgmt.msc`) is the visual way to see everything about a scope at once: the pool, exclusions, leases and reservations.

STEP 1 Check what already holds the address

1. Open **DHCP › <server> › IPv4 › Scope [192.0.2.0] › Address Leases**. Sort by IP address and look for 192.0.2.60. Note the *Unique ID* (MAC) and whether it shows as a reservation.
2. Open **Reservations** in the same scope. If a reservation for 192.0.2.60 exists with a *different* MAC, that is the stale binding.
3. Search the lease list for the new printer's MAC. If it already holds a pool address, write it down; you can reach the printer there during the change.

STEP 2 Back up, then correct or create the reservation

1. Right-click the server › **Backup...**, or export the scope with PowerShell (Method B, Stage 1) so the old state is recoverable.
2. If a stale reservation exists: right-click it › **Properties** and replace the *MAC address* with the new printer's (no separators or with hyphens), or delete it and create a new one.
3. To create: right-click **Reservations** › **New Reservation...**. Enter a *Reservation name* matching the queue (e.g. `LAB-PRN-01`), the IP address, the MAC address, a description with the change number, and supported types *Both* (DHCP and BOOTP; some printers try BOOTP first). Select **Add**.
4. Delete the stale lease: in **Address Leases**, right-click the old entry for 192.0.2.60 (and any pool lease held by the new MAC) › **Delete**.
5. If the scope is in a **DHCP failover** relationship, right-click the scope › **Replicate Scope**. Reservations are not synchronised to the partner automatically.

STEP 3 Make the printer ask again

1. Power-cycle the printer (or reset its network from the front panel). A device only goes through DORA when it boots, renews, or its lease is rejected.
2. Within a minute, refresh **Address Leases**: 192.0.2.60 should show the new MAC with the reservation icon.
3. Print the network configuration label: the printer must report 192.0.2.60, the correct mask and gateway.

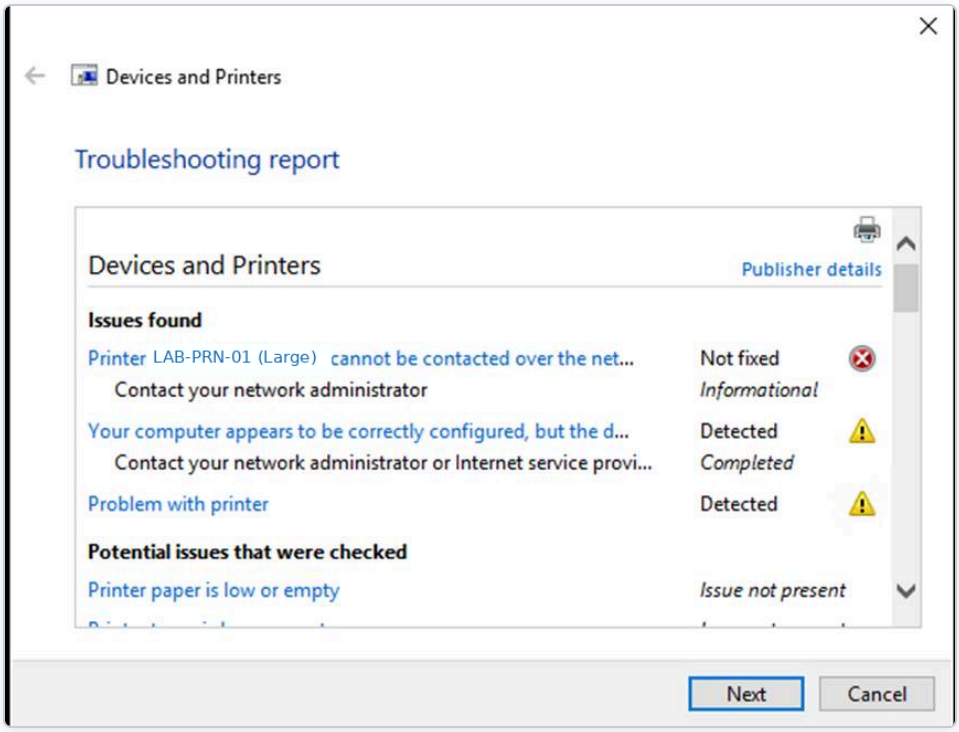


Figure 6. What the stale binding looked like from a workstation: Windows' printer troubleshooter reports the device cannot be contacted although "your computer appears to be correctly configured". The PC is fine; nothing holds the printer's address.

CHAPTER 4

Method B · DhcpServer PowerShell Template

The **DhcpServer** module (installed with the DHCP role or RSAT: `Add-WindowsCapability -Online -Name Rsat.DHCP.Tools~~~~0.0.1.0`) manages every part of a scope remotely. The template below performs the whole reservation change safely: it inspects, exports a backup, removes the stale reservation and leases, adds the new reservation, replicates to a failover partner and writes evidence. Run it with `-WhatIf` first.

SET-PRINTERRESERVATION.PS1 · BASE TEMPLATE POWERSHELL

```powershell
<#
.SYNOPSIS  Bind a printer's MAC address to a fixed IPv4 address on a Windows DHCP server, cleaning stale state.
.DESCRIPTION
  Stage 0  Inspect: scope, reservations and leases for the IP and for the MAC.
  Stage 1  Back up the scope (configuration + leases) to an XML export.
  Stage 2  Remove a reservation/lease that binds the IP to a different MAC, and any pool lease of the new MAC.
  Stage 3  Add the reservation; replicate to the failover partner if the scope is in failover.
  Stage 4  Write JSON evidence. Power-cycle the printer afterwards.
.NOTES     Access: DHCP Administrators on the DHCP server. Requires the DhcpServer module (RSAT).
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string]    $DhcpServer,        # e.g. 'srv-dhcp-01.contoso.com'
  [Parameter(Mandatory)] [ipaddress] $ScopeId,           # e.g. 192.0.2.0
  [Parameter(Mandatory)] [ipaddress] $IPAddress,         # e.g. 192.0.2.60
  [Parameter(Mandatory)] [ValidatePattern('^([0-9A-Fa-f]{2}[-:]?){5}[0-9A-Fa-f]{2}$')] [string] $MacAddress,
  [Parameter(Mandatory)] [string]    $Name,              # e.g. 'LAB-PRN-01'
  [string] $ChangeRef      = 'CHG0000000',
  [string] $EvidenceFolder = 'C:\Temp\DhcpChanges'
)
$ErrorActionPreference = 'Stop'
Import-Module DhcpServer
$mac   = ($MacAddress -replace '[-:]', '' -replace '(..)(?!$)', '$1-').ToUpper()   # 00-11-22-33-44-55
$stamp = Get-Date -Format 'yyyyMMdd_HHmmss'
New-Item -ItemType Directory -Path $EvidenceFolder -Force | Out-Null
$common = @{ ComputerName = $DhcpServer }

# ---- Stage 0: inspect ------------------------------------------------------
$scope   = Get-DhcpServerv4Scope @common -ScopeId $ScopeId
$resIp   = Get-DhcpServerv4Reservation @common -IPAddress $IPAddress -ErrorAction SilentlyContinue
$resMac  = Get-DhcpServerv4Reservation @common -ScopeId $ScopeId -ClientId $mac -ErrorAction SilentlyContinue
$leaseIp = Get-DhcpServerv4Lease @common -IPAddress $IPAddress -ErrorAction SilentlyContinue
$leaseMac= Get-DhcpServerv4Lease @common -ScopeId $ScopeId -ClientId $mac -ErrorAction SilentlyContinue
if ([version]"$IPAddress" -ge [version]"$($scope.StartRange)" -and [version]"$IPAddress" -le [version]"$($scope.EndRange)") {
  Write-Verbose 'Address is inside the scope range; a reservation is valid there, but confirm it is excluded from IPAM pools.'
}
"Before: reservation(IP)=$($resIp.ClientId)  reservation(MAC)=$($resMac.IPAddress)  lease(IP)=$($leaseIp.ClientId)  lease(MAC)=$($leaseMac.IPAddress)"

# ---- Stage 1: back up ------------------------------------------------------
```

```
$backup = Join-Path $EvidenceFolder "$ChangeRef-$stamp-scope.xml"
Export-DhcpServer @common -File $backup -ScopeId $ScopeId -Leases -Force

# ---- Stage 2: clean stale bindings ------------------------------------------
if ($resIp -and $resIp.ClientId -ne $mac -and $PSCmdlet.ShouldProcess("$IPAddress -> $($resIp.ClientId)",
'Remove stale reservation')) {
  Remove-DhcpServerv4Reservation @common -IPAddress $IPAddress
}
if ($resMac -and $resMac.IPAddress -ne $IPAddress -and $PSCmdlet.ShouldProcess("$mac ->
$($resMac.IPAddress)", 'Remove other reservation for this MAC')) {
  Remove-DhcpServerv4Reservation @common -IPAddress $resMac.IPAddress
}
foreach ($l in @($leaseIp, $leaseMac) | Where-Object { $_ }) {
  if ($l.AddressState -notlike '*Reservation*' -and $PSCmdlet.ShouldProcess("$($l.IPAddress)
($($l.ClientId))", 'Remove lease')) {
    Remove-DhcpServerv4Lease @common -IPAddress $l.IPAddress
  }
}

# ---- Stage 3: add reservation and replicate ---------------------------------
if (-not (Get-DhcpServerv4Reservation @common -IPAddress $IPAddress -ErrorAction SilentlyContinue)) {
  if ($PSCmdlet.ShouldProcess("$IPAddress -> $mac", 'Add reservation')) {
    Add-DhcpServerv4Reservation @common -ScopeId $ScopeId -IPAddress $IPAddress -ClientId $mac `
      -Name $Name -Description "Label printer · $ChangeRef" -Type Both
  }
}
$fo = Get-DhcpServerv4Failover @common -ScopeId $ScopeId -ErrorAction SilentlyContinue
if ($fo -and $PSCmdlet.ShouldProcess($fo.PartnerServer, 'Replicate scope to failover partner')) {
  Invoke-DhcpServerv4FailoverReplication @common -ScopeId $ScopeId -Force | Out-Null
}

# ---- Stage 4: evidence ------------------------------------------------------
$after = Get-DhcpServerv4Reservation @common -IPAddress $IPAddress -ErrorAction SilentlyContinue
[pscustomobject]@{
  ChangeRef = $ChangeRef; Operator = "$env:USERDOMAIN\$env:USERNAME"; TimeUtc = (Get-
Date).ToUniversalTime().ToString('o')
  DhcpServer = $DhcpServer; Scope = "$ScopeId"; IPAddress = "$IPAddress"; MacAddress = $mac
  StaleReservationMac = $resIp.ClientId; ReservationAfter = $after.ClientId; FailoverPartner =
$fo.PartnerServer
  Backup = $backup
} | ConvertTo-Json | Out-File (Join-Path $EvidenceFolder "$ChangeRef-$stamp-evidence.json")
Write-Host "Reservation $IPAddress -> $mac in place. Power-cycle the printer, then verify." -
ForegroundColor Green
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION POWERSHELL

```
.\Set-PrinterReservation.ps1 -DhcpServer srv-dhcp-01.contoso.com -ScopeId 192.0.2.0 -IPAddress
192.0.2.60 `
   -MacAddress 00:11:22:33:44:55 -Name 'LAB-PRN-01' -ChangeRef CHG0012345 -WhatIf -Verbose
```

The `-WhatIf` run prints the current bindings and every removal it would make. Re-run without it once the output matches the plan.

4.1 Useful read-only queries

DIAGNOSIS WITH DHCPSERVER CMDLETS POWERSHELL

```powershell
$s = @{ ComputerName = 'srv-dhcp-01.contoso.com' }
Get-DhcpServerv4ScopeStatistics @s -ScopeId 192.0.2.0              # free / in use / % used
Get-DhcpServerv4FreeIPAddress   @s -ScopeId 192.0.2.0 -NumAddress 5 # next free addresses
Get-DhcpServerv4Reservation     @s -ScopeId 192.0.2.0 | Sort-Object IPAddress |
  Format-Table IPAddress, ClientId, Name, Description
Get-DhcpServerv4Lease           @s -ScopeId 192.0.2.0 -BadLeases    # addresses declined as in use
Get-DhcpServerv4Filter          @s                                  # MAC allow/deny lists
```

NOTE · THE DHCP AUDIT LOG

Windows DHCP writes a daily text log at `%windir%\System32\dhcp\DhcpSrvLog-<Day>.log`. Event IDs 10 (new lease), 11 (renewal), 13 (address found in use on the network), 14 (pool exhausted) and 15 (lease denied) tell you whether the printer ever asked and what the server answered. Search it for the printer's MAC without separators.

CHAPTER 5

Methods C and D · Static Address and Queue Integration

5.1 Method C · Static address on the printer

Use a static, on-device address only when there is no DHCP for that network (isolated lab segments, instrument networks) or when the network team requires it. The address must still be **recorded in IPAM and excluded from any DHCP pool**, or DHCP will eventually hand it to something else and create a conflict.

On the printer itself

1. **Front panel**: on Link-OS desktop models open the connection / network settings for the *wired* interface, set *IP protocol* to **Permanent**, then enter IP address, subnet mask and default gateway. Exact menu names vary by model and firmware.
2. **Web page**: browse to the printer's current address, open View and Modify Printer Settings › Network Communications Setup › TCP/IP Settings for the wired print server, sign in with the printer admin password, set the values and `Submit Changes`.
3. **Utility**: the Zebra Setup Utilities / Printer Setup Utility can configure network settings over USB, which is handy when the printer is not yet reachable.
4. Reset the network card or power-cycle so the new settings take effect, then print the network configuration label to confirm.

Scripted, with Set-Get-Do (SGD) commands

Zebra Link-OS printers accept *SGD* commands on the same raw port as print jobs, which makes the change scriptable from the print server while the printer still has its temporary address.

SET-ZEBRASTATICIP.PS1 · SGD OVER TCP 9100 POWERSHELL

```powershell
$current = '192.0.2.143'                    # address the printer has now (pool or old static)
$sgd = @'
! U1 setvar "internal_wired.ip.protocol" "permanent"
! U1 setvar "internal_wired.ip.addr" "192.0.2.60"
! U1 setvar "internal_wired.ip.netmask" "255.255.255.0"
! U1 setvar "internal_wired.ip.gateway" "192.0.2.1"
! U1 do "device.reset" ""
'@
$c = [System.Net.Sockets.TcpClient]::new($current, 9100)
$b = [Text.Encoding]::ASCII.GetBytes($sgd + "`r`n"); $c.GetStream().Write($b, 0, $b.Length); $c.Close()
Start-Sleep 45; Test-Connection 192.0.2.60 -Count 4
```

WATCH OUT · VERIFY THE SGD NAMES FOR YOUR MODEL

The `internal_wired.ip.*` variables apply to the built-in wired print server on Link-OS printers. Wireless and older models use other names. Check the Zebra SGD reference for your firmware before scripting, and keep someone at the printer in case the new settings make it unreachable.

EXAMPLE · REFERENCE CASE 2: A STATIC ADDRESS THAT MOVED

In a later relocation, two printers were moved to new lab rooms. One kept its switch and subnet and worked immediately. The other was patched into a switch in a different subnet; its network configuration showed *IP protocol: Permanent* with an address and gateway from the *old* subnet, so it could not be reached and the operator could not save a new address on the panel. Local IT supplied an address and gateway for the new subnet, the printer was reconfigured, and the queue's port was updated to match. Lesson: a static address belongs to a subnet, not to a printer. Moving it across subnets is a readdressing change, not a cable move.

5.2 Method D · Windows queue, BarTender and LIMS integration

Once the printer reliably holds its address, it must be connected to the systems that print to it. All of these reference the *queue name*, so choose it once and keep it stable.

Create the port and queue with PowerShell

NEW-LABELPRINTERQUEUE.PS1 · ON THE PRINT / BARTENDER SERVER (ELEVATED) POWERSHELL

```powershell
[CmdletBinding(SupportsShouldProcess)]
param(
  [string] $Address    = '192.0.2.60',
  [string] $QueueName  = 'LAB-PRN-01 (Large)',
  [string] $DriverName = 'ZDesigner ZD621-203dpi ZPL',   # must match the printhead resolution
  [string] $Location   = 'Lab bench 3',
  [string] $ChangeRef  = 'CHG0012345'
)
$ErrorActionPreference = 'Stop'
$port = "IP_$Address"
if (-not (Test-NetConnection $Address -Port 9100 -InformationLevel Quiet)) { throw "TCP 9100 on $Address
not reachable." }
if (-not (Get-PrinterPort -Name $port -ErrorAction SilentlyContinue) -and $PSCmdlet.ShouldProcess($port,
'Add-PrinterPort')) {
  Add-PrinterPort -Name $port -PrinterHostAddress $Address -PortNumber 9100
}
if (-not (Get-PrinterDriver -Name $DriverName -ErrorAction SilentlyContinue)) { throw "Install driver
'$DriverName' first." }
if (-not (Get-Printer -Name $QueueName -ErrorAction SilentlyContinue) -and
$PSCmdlet.ShouldProcess($QueueName, 'Add-Printer')) {
  Add-Printer -Name $QueueName -DriverName $DriverName -PortName $port -Location $Location -Comment
"Static/reserved IP · $ChangeRef"
```

```
}
Get-Printer -Name $QueueName | Format-List Name, DriverName, PortName, PrinterStatus
```

If the queue already exists and only the address changed (as in reference case 2), point it at a new port rather than re-creating it: `Add-PrinterPort` for the new address, then `Set-Printer -Name '<queue>' -PortName 'IP_<new address>'`. The queue name, and therefore every BarTender and LIMS reference, is unchanged.

> **TIP · SNMP STATUS AND "OFFLINE" PRINTERS**
>
> Ports created in the GUI have *SNMP Status Enabled* ticked by default. If SNMP is blocked by the firewall or uses a non-default community, Windows may show the queue as *Offline* even though printing works. Either allow UDP 161 with the correct community or untick SNMP on the port (**Printer properties › Ports › Configure Port**).

The GUI equivalent

1. **Print Management › Print Servers › <server> › Ports** › **Add Port...** › *Standard TCP/IP Port*: address 192.0.2.60, port name `IP_192.0.2.60`, device type *Generic Network Card*, raw port 9100.
2. **Printers** › **Add Printer...** › *Add a new printer using an existing port* › choose the port › choose the ZPL driver matching the printhead › name the queue.

BarTender and LIMS

1. In BarTender, open a test format and print to the new queue; confirm the printer appears with the expected driver and stock (**File › Print › Document Properties**).
2. If the site uses BarTender Integration Builder or a print-server service, confirm the integration's printer list includes the new queue, and that the service account can see it.
3. Ask the LIMS administrator to register (or update) the printer record with the exact queue name and, if used, a location/department so it appears in users' printer lists.
4. Print one label from BarTender, then one from the LIMS, with the lab present.

CHAPTER 6

Verification, Escalation and Lessons Learned

6.1 Layer-by-layer verification

VERIFY FROM THE PRINT SERVER AND FROM A LAB PC — POWERSHELL

```
Test-Connection 192.0.2.60 -Count 4                      # ICMP reachability (may be blocked by policy)
Get-NetNeighbor -IPAddress 192.0.2.60 -ErrorAction SilentlyContinue |
  Format-Table IPAddress, LinkLayerAddress, State         # from a PC in the SAME VLAN: MAC must be the printer's
Test-NetConnection 192.0.2.60 -Port 9100                 # TcpTestSucceeded : True is what printing needs
Resolve-DnsName LAB-PRN-01.contoso.com -ErrorAction SilentlyContinue   # if a DNS record was requested
Get-Printer -Name 'LAB-PRN-01 (Large)' | Format-List Name, PortName, PrinterStatus
```

- [ ] Network configuration label shows the assigned IP, mask, gateway and the expected MAC.
- [ ] DHCP shows the reservation with the printer's MAC and an active lease (Methods A/B).
- [ ] ARP from the lab VLAN maps the IP to the printer's MAC; TCP 9100 succeeds from the print server.
- [ ] Test labels from BarTender and the LIMS are correct and accepted by the lab.

6.2 Troubleshooting

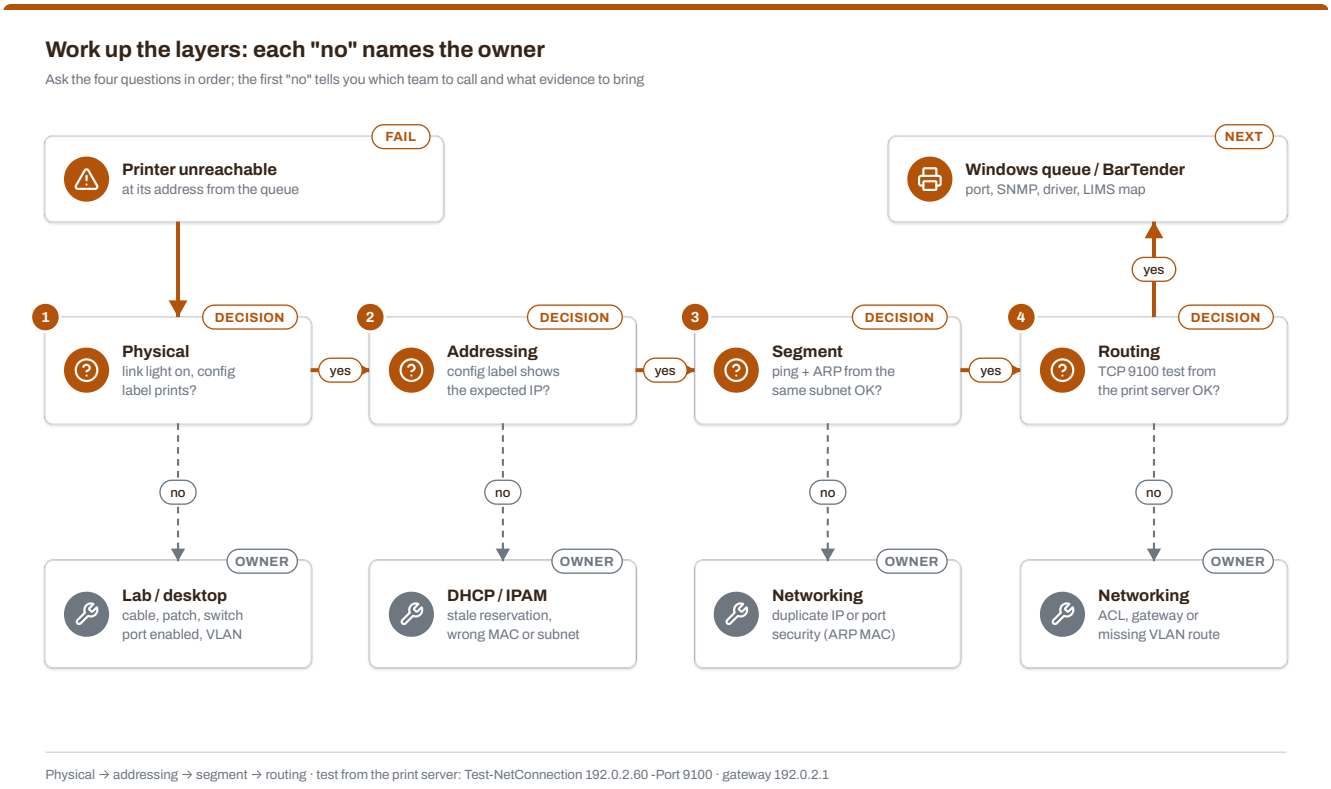


Figure 7. Test the layers in order so the first failing check identifies the owning team and the evidence to hand them.

Symptom	Likely cause and fix
Reservation correct, printer still unreachable	Stale lease still held, or printer has not renewed. Delete the lease, power-cycle the printer, check the DHCP log for event 10 from its MAC.
Config label shows a pool address	Reservation names a different MAC (typo, old printer, wireless vs wired MAC). Rebind to the MAC printed on the label.
Config label shows 0.0.0.0 or a 169.254.x.x address	No DHCP reply: VLAN mismatch, relay missing on the gateway, MAC filter, or pool exhausted (event 14).
ARP shows a different MAC	IP conflict. Find the other device (switch MAC table), move it, and exclude the address from the pool.
Ping fails, TCP 9100 succeeds	ICMP blocked by firewall policy. Printing is fine; record it.
Ping OK, TCP 9100 fails	Firewall/ACL allows ICMP but not 9100, or the printer's raw port is changed. Check the configuration label for the IP port.
Queue shows Offline, jobs print	SNMP status enabled but SNMP blocked. Untick SNMP on the port or open UDP 161.

Printer moved and stopped working	Static address from the old subnet (IP protocol *Permanent*). Readdress for the new subnet, update the queue port.

6.3 Escalating with evidence

Most delay in the reference case came from hand-offs: each team checked its own layer, found nothing wrong, and passed the ticket on. The cure is one ticket as the system of record and an escalation that contains the evidence for every layer you have already cleared, plus precise questions.

ESCALATION EMAIL · TEMPLATE TEXT

```
Subject: [CHG0012345] Label printer LAB-PRN-01 unreachable at reserved address 192.0.2.60 - request
DHCP/ARP check

Hello Networking / IPAM team,

Printer:         Zebra ZD621, queue "LAB-PRN-01 (Large)", lab VLAN 192.0.2.0/24
Expected:        192.0.2.60 reserved for MAC 00-11-22-33-44-55 (from the printer's network config label)
Observed:        No ping, no ARP entry from a PC in the same VLAN; TCP 9100 fails from SRV-PRINT-01.
                 Windows troubleshooter: "device or resource (192.0.2.60) is not responding".

Already verified on our side:
  1. Printer powered, link light on, config label printed (attached).
  2. Reservation 192.0.2.60 -> 00-11-22-33-44-55 present in scope 192.0.2.0 (screenshot attached).
  3. Previous reservation for the same IP (MAC 00-11-22-33-44-01, retired printer) removed at 10:05.
  4. A known-good printer in the same room answers ping and 9100 (so the VLAN is up).

Please could you:
  a) check the DHCP audit log / relay for DISCOVER or REQUEST from 00-11-22-33-44-55 since 10:05;
  b) confirm the ARP / MAC table on the lab gateway and access switch for this MAC;
  c) confirm no ACL or firewall rule blocks SRV-PRINT-01 -> 192.0.2.60 TCP 9100.

On-site contact can power-cycle the printer at any time. Thank you,
<your name> · <team> · <phone>
```

6.4 Lessons learned from the reference engagement

1. **Check the address's history, not just its availability.** Query reservations and leases for both the IP and the MAC first.
2. **Read the MAC from the device.** Spreadsheets and boxes may list the wireless MAC or a previous unit.
3. **Rebinding is three steps:** fix the reservation, delete stale leases, make the printer ask again. Skipping the last two produced the "DHCP delay".
4. **Clear each layer with evidence** (config label, ARP, 9100 test, DHCP log) and attach it to escalations.
5. **A static address belongs to a subnet.** Check addressing on every move; record every assignment in IPAM/CMDB.

CHAPTER 7

Rollback, Glossary and Review

7.1 Rollback

Change	Rollback
Reservation (A/B)	Remove the new reservation and re-add the previous binding from the evidence JSON, or restore the exported scope with `Import-DhcpServer … -ScopeOverwrite`.
Static on printer (C)	Set IP protocol back to DHCP (or *All*) on the panel or web page, or send `! U1 setvar "internal_wired.ip.protocol" "dhcp"` and reset.
Port / queue (D)	`Set-Printer -PortName` back to the previous port; `Remove-Printer` / `Remove-PrinterPort` only for queues that never went live.

7.2 Glossary

DHCP Dynamic Host Configuration Protocol; assigns IP settings to clients.

Reservation Permanent DHCP mapping from a client's MAC address to one IP address.

ARP Address Resolution Protocol; maps IP to MAC on a local segment.

Check your understanding

1. A reservation for 192.0.2.60 exists, but the new printer's configuration label shows 192.0.2.143. What two things are probably wrong, and how do you fix them?
2. Why does a DHCP Discover from a printer in a lab VLAN reach a server in another VLAN, and what field tells the server which scope to use?
3. From a PC in the printer's VLAN, ARP shows 192.0.2.60 with an unexpected MAC. What does that mean, and what is your next step?
4. List the evidence you would attach before escalating an unreachable printer to Networking.
5. Why should an on-device static address still be recorded in IPAM and excluded from the DHCP pool?